OPENCV AI COMPETITION 2026

Northbridge Watch

OpenCV 5 sees the porch & kitchen, then quietly decides: notify, auto-log, or ask a human.

OpenCV 5 | AWS Lambda | S3 | SNS | AWS SAM | Vercel

THE PROBLEM

Home cameras spam alerts. Nobody wants a ping for an empty porch.

Team Northbridge · household of Avery & Morgan Quinn

OUR SOLUTION

Quiet by default.

Perception drives a policy: calm scenes → noop, package/spill → notify, empty fridge → restock log, ambiguity → ask Avery & Morgan.

LIVE DEMO

OPENCV AI COMPETITION 2026 · AGENTIC VISION

Northbridge Watch

OpenCV 5 watches porch and kitchen scenes for Avery & Morgan. Detections drive a quiet agent policy — notify, auto-log, or ask a human.

OpenCV 5 · AWS Lambda · S3 · SNS · Team Northbridge

Demo scenes

Calm porch
Empty Northbridge porch — no action expected.

Package at door
Single brown package on the porch mat.

Kitchen spill
Irregular liquid spill on light tile.

Empty fridge shelf
Mostly empty fridge interior — restock signal.

Busy porch (multi-package)
Several packages blocking the door — notify Avery & Morgan.

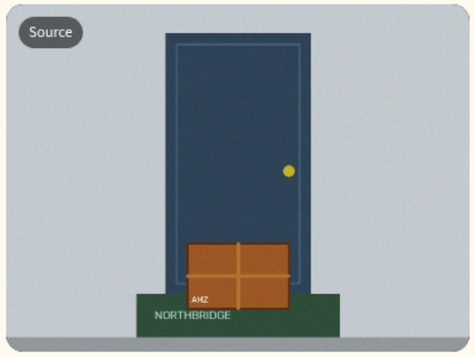
Re-run live OpenCV API

Live API needs `./demo.sh serve`. Vercel uses precomputed traces.

Package at door

Single brown package on the porch mat.

Notify





Perception	package · conf 0.82 · OpenCV 5.0.0
Decision	Single package (conf=0.82).
Speakable	A package arrived on the porch. I can mark it received.

Actions


```
[
  {
    "type": "persist_event",
    "ref": "local://events.jsonl#73ce228a-5a32-4f3f-bb93-3ee5ee91e20a",
    "backend": "local-jsonl"
  },
  {
    "type": "notify",
    "target": "Avery",
    "channel": "sns_or_push_mock",
    "message": "A package arrived on the porch. I can mark it received."
  }
]
```


Northbridge Watch · Apache 2.0 · Pseudonymous household Avery & Morgan · Perception changes the plan — that is the Agentic Vision loop.

HOW IT WORKS

Architecture

- OpenCV 5 detection on scene frames
- Annotated output + structured perception
- Agent policy decides action
- AWS Lambda/S3/SNS for cloud actions; Vercel demo with traces

TRY IT

Northbridge Watch

Live: https://northbridge-watch.vercel.app

Code: github.com/dl88jing/northbridge-watch

OpenCV 5 · AWS Lambda · S3 · SNS · AWS SAM · Vercel